

5. (a) Explain why, for thousands of years, the percentage of carbon dioxide in the Earth's atmosphere remained approximately constant **and** how this stabilised conditions for life on Earth. [6 QER]

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- (b) During the past 200 years, evidence shows that the percentage of carbon dioxide in the atmosphere has been steadily rising.

One factor that contributes to this increase is how people travel to work.

The table shown below is taken from a climate change website.

It gives information about the mass of carbon dioxide produced **each journey per person** travelling to work by different methods each day.

| Distance travelled (km)                          | 10   | 20   | 25   | 50   |
|--------------------------------------------------|------|------|------|------|
| carbon dioxide produced travelling by car (kg)   | 0.60 | 1.20 | 1.50 | 3.00 |
| carbon dioxide produced travelling by bus (kg)   | 0.10 | 0.20 | 0.25 | 0.50 |
| carbon dioxide produced travelling by train (kg) | 0.18 | 0.36 | 0.45 | 0.90 |

Use this information to calculate the mass of carbon dioxide saved during a 5-day working week, if **250** people travel **30 km each way** to work and back home by train instead of every person using their own car.

[4]

mass of carbon dioxide saved = .....kg

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